# THE DUALIZING MODULE OF A COHEN-MACAULAY PROPER SCHEME OVER A FIELD AND ITS SERRE DUALITY PAIRINGS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FVZ (sources/stacks-duality/pinned-original-source.tex line 7106).

*Allowed standard prerequisites (editorial).*

- Duality for a proper scheme over a field. Same chapter, Lemma labelled lemma-duality-proper-over-field, cited at proof lines 7123-7131. Supplies the dualizing complex, the pairing with H^0 and the duality isomorphisms.
- The dualizing module of a Cohen-Macaulay scheme is Cohen-Macaulay. Same chapter, Lemma labelled lemma-dualizing-module-CM-scheme, cited at proof lines 7127-7129. Gives the Cohen-Macaulay property of omega_X and its support.

*Source reference.* Original lemma 0FVV: sources/stacks-duality/pinned-original-source.tex, source lines 6934–6934. Same-chapter pointer into the author environment lemma at source line 6934; invoked at proof line 7124 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original section 0AWH: sources/stacks-duality/pinned-original-source.tex, source lines 6031–6031. Same-chapter pointer into the author environment lemma at source line 6031; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0AWT: sources/stacks-duality/pinned-original-source.tex, source lines 6214–6214. Same-chapter pointer into the author environment lemma at source line 6214; invoked at proof line 7128 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $X$ be a proper scheme over a field $k$ which is Cohen-Macaulay and equidimensional of dimension $d$. The module $\omega_X$ of Lemma 0FVV has the following properties*

(1) *$\omega_X$ is a dualizing module on $X$ (Section 0AWH),*
(2) *$\omega_X$ is a coherent Cohen-Macaulay module whose support is $X$,*
(3) *there are functorial isomorphisms $\mathrm{Ext}^i_X(K, \omega_X[d]) = \mathrm{Hom}_k(H^{-i}(X, K), k)$ compatible with shifts and distinguished triangles for $K \in D_{QCoh}(X)$,*
(4) *there are functorial isomorphisms $\mathrm{Ext}^{d-i}(\mathcal{F}, \omega_X) = \mathrm{Hom}_k(H^i(X, \mathcal{F}), k)$ for $\mathcal{F}$ quasi-coherent on $X$.*

*Proof.* It is clear from Lemma 0FVV that $\omega_X$ is a dualizing module (as it is the left most nonvanishing cohomology sheaf of a dualizing complex). We have $\omega_X^\bullet = \omega_X[d]$ and $\omega_X$ is Cohen-Macaulay as $X$ is Cohen-Macualay, see Lemma 0AWT. The other statements follow from this combined with the corresponding statements of Lemma 0FVV. □